Michael Tin

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Education

University of California, Riverside (UCR) Apr 2026 – Apr 2028
Master of Science in Computer Engineering *Riverside, CA*

University of California, Riverside (UCR) Sep 2021 – Jun 2025
Bachelor of Science in Computer Engineering *Riverside, CA*

Experience

Software Engineering Intern Aug 2026 – Present
National Aeronautics and Space Administration (NASA) *Hampton, VA*

- Develop a comprehensive framework for modernizing legacy PHP code in the NASA Earth Science Strategic Integration Environment (NESSIE) for PHPUnit test and evaluation targeting 100% code coverage.
- Establish a robust CI/CD pipeline and technical documentation within the Flight Systems Software Branch to standardize automated test execution, ensuring the reliability of mission-critical data for Earth science portfolios.

Research Assistant (UIUC SRSE 2026) Jun 2026 – Present
University of Illinois Urbana-Champaign (UIUC) *Remote*

- Develop core framework components for AEBench, a benchmark for evaluating AI agents on artifact evaluation tasks, building a shared runtime to execute versioned research cases and record reproducible results.
- Engineer automated Python evaluation pipelines that compare agent outputs against reference data by utilizing statistical analysis methods, producing objective technical assessments of agent performance.

Technology Director Sep 2022 – Jun 2025
The Highlander Newspaper *Riverside, CA*

- Engineered hardware and network infrastructure for UCR's student-run newspaper, providing on-demand technical support and workflow training to ensure minimal downtime during weekly production.
- Devised a website optimization strategy centered on content delivery optimization, reducing website latency by 82.8% and growing readership by 73.3% to 40,000 monthly average users (MAUs).

Projects

MoodJourney | *Rust, Tauri, Whisper, React* Apr 2025 – Jun 2025

- Developed a cross-platform journal application interfacing with the Google Gemini API, structuring multi-entry prompt context and sentiment histories to generate personalized insights for user self-reflection.
- Engineered an on-device speech-to-text pipeline integrating a local Whisper model within a Rust/Tauri desktop runtime, optimizing local inference execution and memory usage for privacy-focused audio transcription.

XV6 Kernel Optimization | *C, Assembly, RISC-V* Jan 2025 – Mar 2025

- Engineered advanced system calls and symbolic links for the XV6 kernel, expanding core functionality with file system traversal, inter-process communication, and optimized I/O operations.
- Implemented dynamic page fault management techniques, including lazy page allocation, Copy-on-Write (COW) fork, and memory mapping, to streamline program execution while minimizing system resource consumption.

Integrated Arcade System | *C, C++, UART, SPI, AVR* Oct 2024 – Dec 2024

- Engineered a dual-microcontroller system for "Connect 4" and "Whack-a-Mole" using ATmega328P and FRDM-K64F, synchronizing logic via UART/SPI to drive score displays, sound effects, and background music.
- Developed low-level drivers for an ST7735 TFT display, 74HC595 shift registers, and ULN2003 stepper motors, optimizing ISRs to handle asynchronous player inputs while managing game states with physical feedback.

Skills

Programming Languages: Python, C, C++, Rust, Assembly, SQL, Java, MATLAB, Verilog, VHDL, TypeScript
ML Frameworks & Libraries: PyTorch, TensorFlow, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, Whisper
Developer Tools: Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux/WSL, CMake, GDB, QEMU, Valgrind
Embedded Systems & Protocols: ARM, RISC-V, FreeRTOS, STM32, AVR, UART, SPI, I2C, I2S, JTAG, ISR